Identification from restricted measurements

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Abstract

An experiment may identify a sharp interval even when it cannot identify a unique operation. We derive this interval for rectangular quantum operations with a binary unobserved factor. Exact channel and graph examples demonstrate how phase-insensitive measurements lose coherence. A rare-success discrimination problem separates an assignment-independent lower bound, a bound that uses the allocation policy, and an attainable minimum for two specified laws. All trial probabilities in these examples are model values.

1 What a phase scan fixes

In the two-qubit Choi space, use the output-first vectorization $|A\rangle = \mathrm{vec}(A)/2$ for a 4×4 unitary. Write $p = |I_4\rangle$, $u = |iZ \otimes I_2\rangle$ and $v = |iX \otimes I_2\rangle$. They are orthonormal. In a normalized Choi state C write $s = C_{pp}$, $k = C_{uu}$ and $c = C_{pu}$. Read out the projector onto $\sqrt{h}\,p + \sqrt{1-h}\,u$ after applying $U_\phi = P + e^{i\phi}(I - P)$, $P = |p\rangle\langle p|$. The probability is

$$q_\phi = hs + (1-h)k + 2\sqrt{h(1-h)}\,\Re(e^{-i\phi}c).$$

The constant term and the first harmonic determine k and c if s is known and $0 < h < 1$. A coefficient here means the Fourier coefficient of $e^{-i\phi}$, not the peak-to-peak amplitude. For the curve $|\sqrt{sh} - e^{i\phi}\sqrt{k(1-h)}|^2$, $|c|^2 = sk$. The positive block has determinant zero and rank one precisely when $s + k > 0$. This includes the case in which one population vanishes; the zero trace case gives a zero block. Inverting the harmonic amplifies absolute error by $1/\sqrt{h(1-h)}$. No such division is available at either endpoint. In particular, an arbitrary complete phase curve need not saturate the determinant constraint.

Proposition 1. *At each endpoint $h = 0$ or $h = 1$, two trace-preserving channels can agree on the full phase curve and the complete output for each computational-basis input while their Choi purities differ.*

Proof. Take $C_1 = (|p\rangle\langle p| + |u\rangle\langle u|)/4 + |v\rangle\langle v|/2$ and $C_2 = (p-u)(p-u)^\dagger/4 + |v\rangle\langle v|/2$. Their Kraus lists are respectively

$$(I_4/2,\ iZ \otimes I_2/2,\ iX \otimes I_2/\sqrt{2}), \qquad ((I_4 - iZ \otimes I_2)/2,\ iX \otimes I_2/\sqrt{2}).$$

Each list has $\sum K_j^\dagger K_j = I_4$. Both Choi states have $s = k = 1/4$. The endpoint projectors commute with U_ϕ, so both click rates are constantly $1/4$. The nonzero Choi spectra are $(1/2, 1/4, 1/4)$ and $(1/2, 1/2)$, giving purities $3/8$ and $1/2$. For either list, a basis input $|a, b\rangle$ produces the equal mixture of $|a, b\rangle$ and $|1-a, b\rangle$. On $|+, 0\rangle$ the first list gives zero first-qubit Y expectation. In the second, the coherent Kraus term has probability $1/2$ and rotates $+X$ to $+Y$; the other term leaves $+X$. The positive Y outcome has probabilities $1/2$ and $3/4$, respectively. □

2 A graph comparison blind on every pair

For a real symmetric matrix H and either $\psi_{ij}^{\pm} = (|i\rangle \pm i|j\rangle)/\sqrt{2}$, $\langle \psi_{ij}^{\pm}, H\psi_{ij}^{\pm}\rangle = (H_{ii} + H_{jj})/2$. Take the rook and Shrikhande graphs on $(\mathbb{Z}/4\mathbb{Z})^2$. The rook graph connects distinct vertices in a common row or column. The Shrikhande connection set is $\{\pm(1,0), \pm(0,1), \pm(1,1)\}$. Both have Laplacian eigenvalues $0, 4, 8$ with multiplicities $1, 6, 9$. For instance one can evaluate the corresponding Fourier characters, or use the strongly regular identity $A^2 = 4I + 2\mathbf{1}\mathbf{1}^T$ for their adjacency matrices and $L = 6I - A$. Translation invariance makes each spectral projector have constant diagonal equal to its rank divided by 16. Thus each of the 120 labelled pairs has exactly the same three masses $(1, 6, 9)/16$ in both graphs, including the zero-mode output. Every positive-scale probability density calculated from those masses agrees.

The rook graph has eight four-cliques, its rows and columns. A four-clique in the Shrikhande graph would give a triangle among the neighbors of any of its vertices. Those neighbors induce a six-cycle, so there is no four-clique [1]. The graphs are nonisomorphic. By contrast the real plus and minus preparations give $\Delta p_0 = 2/16$. The ratio in the labelled-pair reconstruction theorem then applies under its matched-population and apparatus assumptions. The four-vertex weighted extension that agrees with a rook probe through inverse moment three uses a different comparator and a different probe.

3 A binary discarded factor in any finite first dimension

Consider a nonzero completely positive, trace-nonincreasing branch from $\mathbb{C}^d \otimes \mathbb{C}^2$ to $\mathbb{C}^{d'} \otimes \mathbb{C}^2$. Both first dimensions are positive and can differ. Its Kraus operators are K_j. Write τ_{ab} for the positive first-output marginal on input $|a, b\rangle$ with $1 \le a \le d$, $b = 0, 1$. Its trace s_{ab} is the probability of success. Write $S = \sum s_{ab} > 0$. The local operator subspace is $\mathcal{V} = \{V \otimes I_2 : V : \mathbb{C}^d \to \mathbb{C}^{d'}\}$. Its normalized Choi weight is $w = \sum_j \|\Pi_{\mathcal{V}} K_j\|_F^2 / S$, independent of Kraus representation. Define $F_*(A, B) = \|\sqrt{A}\sqrt{B}\|_1$ and $d_*(A, B) = \operatorname{Tr} A + \operatorname{Tr} B - 2F_*(A, B)$.

Theorem 2. *For prescribed positive τ_{ab} with $0 \le s_{ab} \le 1$ and $S > 0$, the complete set of compatible weights is*

$$[0, w_{\max}], \qquad w_{\max} = 1 - \frac{1}{2S} \sum_{a=1}^{d} d_*(\tau_{a0}, \tau_{a1}).$$

Proof. In second-bit blocks write $K_j = \left(\begin{smallmatrix} A_j & B_j \\ C_j & D_j \end{smallmatrix}\right)$. All four blocks are $d' \times d$. Projection averages the diagonal blocks and removes the others. Consequently $S(1-w) = \sum_j (\|A_j - D_j\|_F^2/2 + \|B_j\|_F^2 + \|C_j\|_F^2)$. For the dilation columns $u_{ab} = \sum_j K_j |a, b\rangle \otimes |j\rangle$, let X_B flip the second output. Then

$$\sum_a \|u_{a0} - X_B u_{a1}\|^2 = \sum_j (\|A_j - D_j\|_F^2 + \|C_j - B_j\|_F^2) \le 2S(1 - w).$$

Uhlmann's purification theorem bounds the real overlap of each aligned pair by F_* of its two marginals. This bounds the left side below by the sum of the d_*'s [2, Theorem 3.22].

For each a choose purifications attaining this overlap in an environment orthogonal to those used for the other first labels. Give each column its second output flag $|b\rangle$.

Different columns are mutually orthogonal with squared norms s_{ab}. They define a single contraction, hence a common admissible branch. Its off-diagonal blocks vanish, and every pair attains equality simultaneously. A zero marginal is represented by a zero column. Now flip the discarded output of this attaining branch. All Kraus blocks are off diagonal in the second factor, so their projections vanish. Tracing out that factor leaves the first-output map identical for every input. A convex mixture with the first branch has the same S and interpolates linearly in w. This proves the complete interval. □

For a larger discarded dimension, an optimization over positive Choi matrices with prescribed marginal constraints is available directly. Multiple overlaps for one first label may require common purifications. The independent-pair argument above provides no closed formula in that regime.

4 Discrimination with rare successful reports

Fix two simple independent branches and a common two-outcome detector. Success has probability h on every input. In the null law every successful first output occupies the high eigenspace. In the alternative, even inputs still occupy the high eigenspace and odd inputs occupy the low eigenspace. Their maximum compatible weights are 1 and $1/2$. The per-success click probabilities on odd inputs are U in the null and L in the alternative, with $0 < L < U < 1$. Let q bound the probability of choosing an odd input at every history. A private policy may depend on earlier observations.

Proposition 3. *For any decision by horizon N of null size at most α, its alternative rejection probability is at most* $\min\{1, \alpha + 1 - (1 - qh(U - L))^N\}$*. If it also attains power at least $1 - \beta > \alpha$, then*

$$Nqh\, D_{\mathrm{Ber}}(L\|U) \geq D_{\mathrm{Ber}}(1 - \beta\|\alpha).$$

Here $D_{\mathrm{Ber}}(p\|r) = p \log(p/r) + (1 - p) \log[(1 - p)/(1 - r)]$*.*

Proof. While the two pasts agree, draw the same assignment from the specified policy in both experiments. The two conditional laws agree on even assignments and have total variation $h(U-L)$ on odd assignments. A maximal coupling keeps the next records equal with probability at least $1-qh(U-L)$. Iteration gives the stated power bound. For relative entropy take the alternative as the first law. The conditional assignment probabilities cancel in the likelihood ratio. A distinguishing input contributes $hD_{\mathrm{Ber}}(L\|U)$ and an even input contributes zero. The chain rule bounds path relative entropy above by the left side. Data processing to the binary decision bounds it below by the right side, using monotonicity for power $> \alpha$ [3]. □

For $h = 0.001$, $(L, U) = (0.2, 0.8)$ and $\alpha = \beta = 0.05$, coupling gives necessary counts 3,837, 5,116, 7,675 and 15,350 for $q = 1, 0.75, 0.5, 0.25$, respectively. With $(L, U) = (0.45, 0.55)$ and $q = 0.5$, the relative-entropy bound requires 264,114 attempts. These inequalities do not construct a procedure that attains the stated error levels.

Exact threshold for two specified laws

Take $(L, U) = (1/5, 4/5)$. Under always-odd assignment, let M be the number of successes among N attempts. Under uniform independent four-input assignment, count only successful odd assignments. In these designs $M \sim \mathrm{Bin}(N, t)$ with $t = h$ and $h/2$, respectively.

Conditional on $M = m$, the number K of informative no-clicks is binomial with parameter $1/5$ under the null, $4/5$ under the alternative. All other observations have likelihood ratio one. The full likelihood ratio is 4^{2K-m}. The optimal equal-prior rule selects the alternative if $2K > m$ and randomizes fairly at equality. Symmetry makes its two errors equal. Therefore 5% size and 95% power are possible for these two laws exactly when the optimal average error is at most $1/20$.

Write $e_m = \Pr_{1/5}(K > m/2) + \frac{1}{2}\Pr_{1/5}(K = m/2)$. The risk is $\sum_{m=0}^{N} \binom{N}{m} t^m (1-t)^{N-m} e_m$. The verifier evaluates the first 66 terms by integer arithmetic and bounds the omitted risk by half the omitted binomial probability. The width is below 10^{-38} at each of the four reported counts. The exact comparisons with $1/20$ give the following decimal summaries.

Design	Attempts	Minimum average error
Always odd	7,064	0.05001038431917
Always odd	7,065	0.04999782557675
Uniform independent	14,130	0.05000391973171
Uniform independent	14,131	0.04999764112286

An extra observation can be discarded, so optimal risk is nonincreasing. Always-odd observations can simulate any adaptive policy for this pair: keep the next observation for an odd assignment or generate an independent draw from the common even law. Thus 7,065 is necessary for any such policy and sufficient for the always-odd comparison. The 14,131 threshold concerns the specified uniform policy. Neither number is a sufficient sample size for the composite four-arm sequential test. Ville's inequality controls that test's false-rejection probability at any stopping time under the stated conditional null [4].

5 Reading the identification map quantitatively

The map separates a mathematical implication, an instrument premise, and a prospective test. Several observations in the same panel answer different questions. The graph moment and instrument results below illustrate the distinction.

- For $n = 16$ the ratio method uses two preparations on each of 120 pairs. Common observation windows give entry bias at most $B/2$ for the omitted positive first-moment operator of norm at most B. A protocol without a zero-mode detector requires a separate common-contrast assumption and additional moments. Matching a consistency identity cannot calibrate a nonlinear change of scale.

- In the specified coherent spectral design the two monitored outputs yield Fisher coefficient $0.4780676701\ldots$ and access to all three yields $0.8279358532\ldots$. At the optimal equal scales, eigenvalue coordinates $2.5212815658\ldots$ and $0.2637461879\ldots$ give the same full output. Local precision and global identification are distinct.

- The finite graph moment witness is a labelled rook edge and a weighted graph. Their shared inverse moments through order three are $1, 5/32, 7/256, 11/2048$. Their fourth moments are $19/16384$ and approximately 0.0068045648751154. The gap $27/2{,}945{,}024$ belongs to a separate pair of positive spectral measures. For those measures 57,291,466 trials suffice under the specified one-sided variance calculation. That budget does not transfer to either graph comparison.

- For the rook-edge versus Shrikhande-nonedge counting protocols, the fixed threshold $\log 2/4$ has equal-prior 10% error crossings of 181 and 188 for equal and antisymmetric phase preparations. Optimizing a repeated upper-threshold bit gives 165 and 173. Full scale records give numerical crossings of 117 and 127; those are not certified optimal experimental budgets. The antisymmetric sample mean has a sufficient 375-trial guarantee for each error. The parity, phase and rare-success tests use different outcomes and nulls.
- In the interior fixed-detector phase experiment with a saturating curve, the three-axis Choi-purity bound uses coefficients $91/1230$ and $7/97$ for longitudinal and transverse squared deviations. At exact two-dimensional support a transferred transverse measurement can determine the remaining coherence. At positive leakage, first-qubit measurements can miss a relative sign detectable by second-qubit Y in the exhibited family. These conclusions depend on the trace-preserving marginal and its transfer to a product preparation.
- Balanced phase offsets illustrate what must transfer across arms. In the specified general density-state model at $\pm 60^\circ$, a trace-distance cap below $0.830970598587\ldots$ guarantees a strict increase, and at that critical cap general states can attain zero. Normalized trace-preserving Choi states still give a strict increase at the cap. The unrestricted trace-preserving sign at $\pm 60^\circ$ remains open; a unitary pair gives a decrease at $\pm 61^\circ$. These results require the same detector and the stated calibration on the conditional preparations.

These calculations use supplied quantum states, tensor factors and labelled ports. Independent evidence of a source, phase operation, common detector and trial accounting determines whether a physical device instantiates a particular observation model.

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